REDIS

What is Redis?

Redis stands for Remote Dictionary Server. It is an in-memory data store. This means Redis stores all data in RAM (Random Access Memory) instead of on hard disk like PostgreSQL.

PostgreSQL = Filing Cabinet
- Stores data on disk (hard drive)
- Slow to access (milliseconds)
- Persistent (data stays after restart)
- Good for: User accounts, orders, transactions

Redis = Whiteboard
- Stores data in RAM (memory)
- Very fast to access (microseconds)
- Temporary (data can expire)
- Good for: Cache, sessions, real-time data

Feature

- In-Memory:All data stored in RAM for speed
- Key-Value Store: Data stored as key-value pairs
- Single Threaded :One thread handles all operations

What Redis Is Used For

Caching

This is Redis's primary role.
Example: an e-commerce app You request a product:

GET /products/1

Instead of hitting the database every time:
First request → fetch from DB and store in Redis
Next requests → serve directly from Redis

Storing User Sessions

When users log in, session data can be stored in Redis:
Why Redis works well here:

- fast access;
- built-in expiration (TTL);
- automatic cleanup of old sessions.

Pub/Sub (Real-Time Messaging)

Redis supports a publish/subscribe model.
Example: a chat app

- A user sends a message;
- Redis publishes it to a channel;
- All subscribers receive it instantly.

User A → Redis channel → User B, User C

This is commonly used for chats, notifications, and live updates.

Queues and Background Jobs

Redis can act as a lightweight task queue.
Example:

- A user uploads an image;
- A job is added to Redis;
- A worker processes it asynchronously.

```
queue:images -> ["img1.jpg", "img2.jpg"]
```

Benefits:

- offloads work from the main app;
- enables async processing.

Counters and Analytics

Redis is great for tracking counters:

```
page:home:views -> 10234
```

Increment operation:

```
INCR page:home:views
```

Used for: likes, views, metrics and analytics

- Every request hits database
- Database gets slow with many requests
- Same data fetched repeatedly

Working

Redis works as a caching layer between the client and the main database to provide faster data access and reduce database load.

• **Request Handling:** When a client requests data, the API gateway first checks Redis for the requested data.

• **Fast Data Retrieval:** If the data is available in Redis (cache hit), it is returned instantly without querying the database.
• **Database Fallback:** If the data is not found in Redis (cache miss), the system fetches it from the main database and stores it in Redis for future requests.

Example: When a user opens a frequently visited product page on an e-commerce site, Redis quickly returns the cached product details instead of querying the database every time

```
User Request → FastAPI → Redis Cache → Return Response
                        |
                        ▼ (if not in cache)
                  PostgreSQL → Cache in Redis → Return
```

Benefits:
- Most requests served from cache
- Database load reduced